

CROSS PRODUCT & CURL (determinant)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$
$$(A_y B_z - A_z B_y)\hat{x} - (A_x B_z - A_z B_x)\hat{y} + (A_x B_y - A_y B_x)\hat{z}$$
$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z row survives.

DOT PRODUCTS $\hat{a} \cdot \hat{b}$

Same dir = 1; perpendicular = 0.						
	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$	$\hat{\theta}$
$\hat{x}$	1	0	0	0	0	0
$\hat{y}$	0	1	0	0	0	0
$\hat{z}$	0	0	1	0	0	0
$\hat{\phi}$	0	0	0	0	1	0

SURFACE ELEMENT $d\mathbf{s}$

$d\mathbf{s} = \hat{n} da$ RHR: curl fingers in loop dir. $\Rightarrow$ thumb = $\hat{n}$	
Loop plane	$d\mathbf{s}$
x-y plane	$\hat{z} dx dy$ or $\hat{z} r dr d\phi$
y-z plane	$\hat{x} dy dz$
x-z plane	$\hat{y} dx dz$ (note $\hat{y}$, not $-\hat{y}$!)
toroid/coplanar	$\hat{\phi} dr dz$
Uniform $\mathbf{B}$: $\iint d\mathbf{a} = A$ (just loop area). RHR picks sign	

B FIELDS

Infinite wire *single wire carrying I*

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}$$

Inside toroid *N-turn coil carrying I*

$$\mathbf{B} = \hat{\phi} \frac{\mu N I}{2\pi r}, \quad \mu = \mu_r \mu_0$$

N in B only if source is multi-turn coil. Single wire: N = 1 (drop it). N in V_{emf} = -NdΦ/dt is the receiver turns.

∂B/∂t TABLE

B(t)	∂B/∂t
B ₀ sin(ωt)	ωB ₀ cos(ωt)
B ₀ cos(ωt)	-ωB ₀ sin(ωt)
B ₀ e ^{-αt}	-αB ₀ e ^{-αt}
B ₀ (const)	0

SIGN → CURRENT DIRECTION

V _{emf}	Inside source	I dir.
< 0	- → + terminal	- $\hat{\phi}$ (CW)
> 0	+ → - terminal	+ $\hat{\phi}$ (CCW)
= 0	—	none

Lenz: induced I opposes the change in Φ.

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a \rightarrow b$

$$\int_a^b \frac{dr}{r} = \ln \frac{b}{a} = \ln b - \ln a$$

$\ln 2 = 0.693; \ln 3 = 1.099; \ln 1.2 = 0.182; \ln 1.5 = 0.405$

j IDENTITIES

Expr	Val	Expr	Val	Expr	Val
j^2	-1	$1/j$	-j	j^3	-j
$e^{j\pi/2}$	j	$e^{-j\pi/2}$	-j	$e^{j\pi}$	-1
$ a + jb $	$\sqrt{a^2 + b^2}$	$\angle(a + jb)$	$\arctan(b/a)$	$j \cdot (-j)$	1

$\text{Re}\{Ae^{j\theta}e^{j\omega t}\} = A\cos(\omega t + \theta).$

UNIT CIRCLE (KEY ANGLES)

θ	$\cos \theta$	$\sin \theta$	θ	$\cos \theta$	$\sin \theta$
0	1	0	$2\pi/3$	-1/2	$\sqrt{3}/2$
$\pi/6$	$\sqrt{3}/2$	1/2	$3\pi/4$	$-\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	π	-1	0
$\pi/3$	1/2	$\sqrt{3}/2$	$3\pi/2$	0	-1
$\pi/2$	0	1	2π	1	0

SI PREFIXES

Prefix	Sym	Val	Prefix	Sym	Val
pico	p	10 ⁻¹²	kilo	k	10 ³
nano	n	10 ⁻⁹	mega	M	10 ⁶
micro	μ	10 ⁻⁶	giga	G	10 ⁹
milli	m	10 ⁻³	tera	T	10 ¹²

EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}} = - \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *loop moves, B static*

$$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = - \frac{d\Phi}{dt}, \quad \Phi = \iint_S \mathbf{B} \cdot d\mathbf{s}$$

N-turn coil

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt}, \quad \lambda = N\Phi$$

Uniform B shortcut *B same everywhere on loop*

$$V_{\text{emf}} = - \frac{\partial B}{\partial t} \cdot A$$

Long wire + rect loop *wire near loop, r: a → b, height c*

$$V_{\text{emf}} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz$$

ROTATING LOOP — GENERATOR

$$\Phi = AB_0 \cos(\omega t + \phi_0), \quad V_{\text{emf}} = AB_0 \omega \sin(\omega t + \phi_0)$$

$$V_0 = N A \omega B_0 \text{ (peak)}, \quad \omega = 2\pi f = 2\pi \frac{\text{rpm}}{60}$$

2nd method: $V_{\text{emf}} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$ on moving sides only.

IDEAL TRANSFORMER

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}, \quad Z_{\text{in}} = \left(\frac{N_1}{N_2}\right)^2 Z_L$$

Step-up ($N_2 > N_1$): $V_2 > V_1, I_2 < I_1$. Power conserved: $V_1 I_1 = V_2 I_2$.

EQUIVALENT CIRCUIT (INDUCTOR IN CHANGING B)

Inductor (resistance R_i) in external changing $\mathbf{B}$, connected to load R :

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i} \approx \frac{V_{\text{emf}}^{\text{tr}}}{R} \text{ when } R_i \ll R$$

$V_{\text{emf}}^{\text{tr}}$ acts as a voltage source in series with R_i .

CROSS PROD CYCLE

$$\hat{x} \times \hat{y} = \hat{z}; \hat{y} \times \hat{z} = \hat{x}; \hat{z} \times \hat{x} = \hat{y}$$
$$\hat{y} \times \hat{x} = -\hat{z}; \text{reverse: negate}$$
$$\mathbf{a} \times \mathbf{a} = 0; |\mathbf{a} \times \mathbf{b}| = ab \sin \theta$$

B FIELD QUICK

Wire: $B = \mu_0 I / (2\pi r)$
Toroid: $B = \mu N I / (2\pi r)$
 $\Phi = \iint \mathbf{B} \cdot d\mathbf{s}$ (Wb)
 $V_{\text{emf}} = -Nd\Phi/dt$

GENERATOR

$$\Phi = AB_0 \cos(\omega t + \phi_0)$$
$$V_{\text{emf}} = AB_0 \omega \sin(\omega t + \phi_0)$$
$$V_0 = N A \omega B_0 \text{ (peak)}$$
$$V_{\text{in}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

j QUICK

$$j^2 = -1; 1/j = -j$$
$$e^{\pm j\pi/2} = \pm j; e^{j\pi} = -1$$
$$|a + jb| = \sqrt{a^2 + b^2}$$
$$\text{Re}\{Ae^{j(\omega t + \theta)}\} = A\cos(\omega t + \theta)$$

CONSTANTS

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$
$$\epsilon_0 \approx \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$$
$$c = 3 \times 10^8 \text{ m/s}$$
$$\eta_0 \approx 377 \approx 120\pi \Omega$$

MAXWELL'S EQUATIONS

Law	Differential	Integral
Gauss E	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint \mathbf{D} \cdot d\mathbf{s} = Q$
Gauss B	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{s} = 0$
Faraday	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint \mathbf{H} \cdot d\mathbf{l} = I_c + I_d$

$\mathbf{D} = \epsilon \mathbf{E}; \mathbf{B} = \mu \mathbf{H}; \mathbf{J} = \sigma \mathbf{E}$. *Phasor:* $\epsilon_c = \epsilon' - j\sigma/\omega$.

BOUNDARY CONDITIONS

Field	General	PEC ($\sigma \rightarrow \infty$)
Tang. E	$E_{t1} = E_{t2}$	$E_t = 0$
Tang. H	$H_{t1} - H_{t2} = J_s$	$\hat{n}_2 \times \mathbf{H} = \mathbf{J}_s$
Norm. D	$D_{n1} - D_{n2} = \rho_s$	$D_n = \rho_s$
Norm. B	$B_{n1} = B_{n2}$	$B_n = 0$

No J_s , no ρ_s : all four components continuous across boundary.

CONDUCTION vs DISPLACEMENT CURRENT

Conduction <i>resistive, $\sigma \neq 0$</i>
$E = \frac{V(t)}{d}, \quad J_c = \sigma E, \quad I_c = \frac{\sigma A}{d} V(t) = \frac{V}{R}, \quad R = \frac{d}{\sigma A}$
Displacement <i>capacitive, changing E</i>
$J_d = \epsilon \frac{\partial E}{\partial t}, \quad I_d = \frac{\epsilon_r \epsilon_0 A}{d} \frac{dV}{dt} = C \frac{dV}{dt} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d}$
Lossy capacitor ($R \parallel C$) <i>both σ and ϵ_r</i>
$I = I_c + I_d = \frac{V}{R} + C \frac{dV}{dt}$

Phasor: $\tilde{J}_d = j\omega\epsilon\tilde{E}$. *In good conductor $I_d \ll I_c$.*

CHARGE RELAXATION

$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \Rightarrow \oint \mathbf{J} \cdot d\mathbf{s} = 0$ (KCL)
$\rho_v(t) = \rho_{v0} e^{-t/\tau_r}, \quad \tau_r = \frac{\epsilon}{\sigma} = \frac{\epsilon_r \epsilon_0}{\sigma}$ (s)
Find σ from decay data
$\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \epsilon_r \epsilon_0}{t_1}$

Cu: $\tau_r \approx 10^{-19}$ s. *Mica:* $\tau_r \approx 10^4$ s. *Distilled water:* $\sim 10^{-6}$ s.

PHASOR METHOD ($\tilde{\mathbf{E}} \rightarrow \tilde{\mathbf{H}}$)

1. $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z)e^{j\omega t}\}$ strip $e^{j\omega t}$, keep e^{-jkz}
2. Time $\rightarrow$ phasor: $\cos(\omega t - kz) \rightarrow e^{-jkz}; \quad \cos(\omega t - kz + \phi) \rightarrow e^{j\phi}e^{-jkz}$ $\sin(\omega t - kz) \rightarrow -je^{-jkz}; \quad \partial/\partial t \rightarrow j\omega$
3. Faraday in phasor domain (plane wave: only ∂_z survives) $\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}$
4. Phasor $\rightarrow$ time: $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}}e^{j\omega t}\}$ $\text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$ $\text{Re}\{-je^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

DYNAMIC POTENTIALS

$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$
<i>Statics:</i> $\mathbf{E} = -\nabla V$. <i>Dynamic:</i> must include $-\partial \mathbf{A}/\partial t$ term. <i>Retarded:</i> effects at $\mathbf{R}$ at time t depend on source at time $t - R'/u_p$.
LOSSLESS PLANE WAVE ★ <i>(missing from CS1+CS2)</i>

LOSSLESS PLANE WAVE ★ (missing from CS1+CS2)

Wavenumber & impedance

$$k = \omega \sqrt{\mu \epsilon} \text{ (rad/m)}, \quad \eta = \sqrt{\frac{\mu}{\epsilon}} \text{ (}\Omega\text{)}$$

Phase velocity & wavelength

$$u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu \epsilon}} \text{ (m/s)}, \quad \lambda = \frac{2\pi}{k} = \frac{u_p}{f} \text{ (m)}$$

Phasor fields (wave in $+\hat{z}$, $\mathbf{E}$ in $\hat{x}$)

$$\tilde{\mathbf{E}}(z) = E_{x0}^+ e^{-jkz} \hat{x}, \quad \tilde{\mathbf{H}}(z) = \frac{E_{x0}^+}{\eta} e^{-jkz} \hat{y}$$

Real fields

$$\mathbf{E}(z, t) = |E_{x0}^+| \cos(\omega t - kz + \phi^+) \hat{x}$$

$$\mathbf{H}(z, t) = \frac{|E_{x0}^+|}{\eta} \cos(\omega t - kz + \phi^+) \hat{y}$$

TEM: $\mathbf{E} \perp \mathbf{H} \perp \hat{k}$. Lossless: $\mathbf{E}$ and $\mathbf{H}$ are in phase.

DIRECTION RELATION ★ (missing from CS1+CS2)

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\mathbf{E}}, \quad \tilde{\mathbf{E}} = -\eta \hat{k} \times \tilde{\mathbf{H}}$$

$\hat{k}$	$\tilde{E}$	$\tilde{H} = \hat{k} \times \tilde{E}$	Example $\tilde{\mathbf{E}}$
$+\hat{z}$	$+\hat{x}$	$+\hat{y}$	$E_0 e^{-jkz} \hat{x}$
$+\hat{z}$	$+\hat{y}$	$-\hat{x}$	$E_0 e^{-jkz} \hat{y}$
$-\hat{z}$	$+\hat{x}$	$-\hat{y}$	$E_0 e^{+jkz} \hat{x}$
$+\hat{x}$	$+\hat{y}$	$+\hat{z}$	$E_0 e^{-jkx} \hat{y}$

POLARIZATION ★ *(missing from CS1+CS2)*

General phasor $a_x = |E_{x0}|$, $a_y = |E_{y0}|$, $\delta = \text{phase } E_y \text{ rel. } E_x$

$$\tilde{\mathbf{E}} = (a_x \hat{x} + a_y e^{j\delta} \hat{y}) e^{-jkz}$$

Type	Condition	Form
Linear	$\delta = 0, \pi$	$(a_x \hat{x} \pm a_y \hat{y}) \cos(\omega t - kz)$
LHC	$a_x = a_y$, $\delta = +\pi/2$	$\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y}) e^{-jkz}$
RHC	$a_x = a_y$, $\delta = -\pi/2$	$\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y}) e^{-jkz}$
Elliptical	general a_x, a_y, δ	traces ellipse in x - y

LHC real field

$\mathbf{E} = a \cos(\omega t - kz)\hat{x} - a \sin(\omega t - kz)\hat{y}$

RHC real field

$\mathbf{E} = a \cos(\omega t - kz)\hat{x} + a \sin(\omega t - kz)\hat{y}$

Elliptical axial ratio $a_\xi = \text{semi-major}, a_\eta = \text{semi-minor}$

$\tan 2\gamma = (\tan 2\psi_0) \cos \delta, \quad \sin 2\chi = (\sin 2\psi_0) \sin \delta$

$\psi_0 = \tan^{-1}(a_y/a_x), \quad R = a_\xi/a_\eta, \quad \tan \chi = \pm 1/R$

$\tan \chi > 0$: LH. $\tan \chi < 0$: RH. $R = 1$: circular. $a_\eta = 0$: linear.

MEDIUM TYPES ★ *(missing from CS1+CS2)*

Type	ϵ''/ϵ'	Key approx.
Low-loss diel.	$\ll 1$	$\alpha \approx \frac{\sigma}{2} \sqrt{\mu/\epsilon'}, \beta \approx \omega \sqrt{\mu\epsilon'}$
Quasi-conductor	10^{-2}	use general α, β
Good conductor	$\gg 100$	$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$

$\delta_s = \frac{1}{\alpha}$ (m), $\delta_s^{\text{cond}} = \frac{1}{\sqrt{\pi f \mu \sigma}}$ (m), $\epsilon'' = \frac{\sigma}{\omega}$

POYNTING & POWER

$\mathbf{S}_{\text{av}} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$ (W/m ²)
Lossless $\alpha = 0, \eta$ real
$\mathbf{S}_{\text{av}} = \frac{ E_0 ^2}{2\eta} \hat{k}$
Lossy $\eta_c = \eta_c e^{j\theta_\eta}$, wave in $+\hat{z}$
$\mathbf{S}_{\text{av}} = \frac{ \tilde{E}(0) ^2}{2 \eta_c } e^{-2\alpha z} \cos \theta_\eta \hat{z}$
<i>Power $\propto e^{-2\alpha z}$, field $\propto e^{-\alpha z}$: power decays at $2\times$ rate.</i>
ATTENUATION & dB

α (dB/m) = 8.69 α (Np/m)
$A = 10 \log \left(\frac{S(z)}{S(0)} \right) = -8.69 \alpha z$ (dB)
$A = 20 \log \left(\frac{ E(z) }{ E(0) } \right)$ (dB)
$G = 10 \log \left(\frac{P_1}{P_2} \right) = 20 \log \left(\frac{V_1}{V_2} \right)$ (dB)
$P \times 2 \Rightarrow +3 \text{ dB}; P \times 10 \Rightarrow +10 \text{ dB}; E \times 2 \Rightarrow +6 \text{ dB}$.

MAXWELL PHASORS

$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$
 $\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}$
 $\epsilon_c = \epsilon' - j\sigma/\omega$
 $\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}$

WAVE PARAMS

$k = \omega\sqrt{\mu\epsilon}$ (rad/m)
 $\eta = \sqrt{\mu/\epsilon}$ (Ω)
 $u_p = 1/\sqrt{\mu\epsilon}$ (m/s)
 $\lambda = 2\pi/k = u_p/f$ (m)

POLARIZATION

Linear: $\delta = 0$ or π
LHC: $a_x = a_y, \delta = +\pi/2$
RHC: $a_x = a_y, \delta = -\pi/2$
Ellip.: general a_x, a_y, δ

MEDIUM + SKIN

$\epsilon_c = \epsilon' - j\sigma/\omega$
 $\delta_s = 1/\alpha$ (general)
RHC: $a_x = a_y, \delta = -\pi/2$
Ellip.: general a_x, a_y, δ
 $R_s = \sqrt{\pi f \mu / \sigma}$ (Ω)

dB QUICK

$\alpha_{\text{dB}} = 8.69 \alpha_{\text{Np}}$
 $G = 10 \log(P_1/P_2)$
 $A = -8.69 \alpha z$ (dB)
 $A = 20 \log(|E(z)|/|E(0)|)$

LOSSY MEDIUM: EXACT GENERAL FORMULAS ★

Complex permittivity
$\varepsilon_c = \varepsilon' - j\varepsilon'' = \varepsilon_r \varepsilon_0 - j \frac{\sigma}{\omega}, \quad \varepsilon'' = \frac{\sigma}{\omega}$
Propagation constant $\gamma^2 = -\omega^2 \mu \varepsilon_c$
$\gamma = \alpha + j\beta$
Exact α (attenuation) Np/m
$\alpha = \omega \sqrt{\frac{\mu \varepsilon'}{2}} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} - 1 \right]^{\frac{1}{2}}$
Exact β (phase constant) rad/m
$\beta = \omega \sqrt{\frac{\mu \varepsilon'}{2}} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} + 1 \right]^{\frac{1}{2}}$
Complex intrinsic impedance
$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \eta_c e^{j\theta_\eta}, \quad \theta_\eta \in \left[0, \frac{\pi}{4} \right]$
Lossy phasor fields $wave\ in\ +\hat{z}, \mathbf{E}\ \parallel\ \hat{x}$
$\tilde{\mathbf{E}}(z) = E_0 e^{-\alpha z} e^{-j\beta z} \hat{x}, \quad \tilde{\mathbf{H}}(z) = \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{y}$
E and H out of phase by θ_η in lossy medium (unlike lossless).
Real fields: $\mathbf{H}(z, t) = \frac{ E_0 }{ \eta_c } e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \hat{y}$

CURRENT IN GOOD CONDUCTOR ★

AC current flows near surface in layer $\sim \delta_s$; DC current uniform.
Current density at depth z $J_0 = \sigma E_0$, surface at $z = 0$
$\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \hat{x} \text{ (A/m}^2\text{)}$
Total current, slab width w (y -dir, integrate z : $0 \rightarrow \infty$)
$\tilde{I} = \frac{J_0 w \delta_s}{1+j} \text{ (A)}$
Surface impedance $per\ square$
$Z_s = \frac{1+j}{\sigma \delta_s} = R_s(1+j), \quad R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \text{ (}\Omega\text{)}$
AC resistance, slab (width w , length ℓ)
$R_{ac} = R_s \frac{\ell}{w} \text{ (}\Omega\text{)}$
Coaxial AC resistance $inner\ a, outer\ b$
$R'_{coax} = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \text{ (}\Omega/m\text{)}$
Power dissipated: $P = \frac{1}{2} I ^2 R_s (\ell/w)$.
$R_s \propto \sqrt{f}$: higher frequency $\Rightarrow$ more loss, smaller δ_s .

READ-A-WAVE: LOSSLESS

Given $\tilde{\mathbf{E}}(z, t) = E_0 \cos(\omega t - kz + \phi) \hat{a}$:
1. Read $\omega, k, E_0, \phi, \hat{a}$; $\hat{k} = +\hat{z}$ (from $-kz$)
2. $f = \omega/(2\pi)$; $\lambda = 2\pi/k$; $u_p = \omega/k$
3. $\eta = \eta_0 \sqrt{\mu_r/\varepsilon_r}$ (real)
4. Phasor: $\tilde{\mathbf{E}} = E_0 e^{j\phi} e^{-jkz} \hat{a}$
5. $\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\mathbf{E}}$; $\mathbf{H} = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$
6. $S_{av} = E_0^2/(2\eta)$ (W/m ²)
READ-A-WAVE: LOSSY
Given $\tilde{\mathbf{E}}(z) = E_0 e^{-\alpha z} e^{-j\beta z} \hat{x}$:
1. Compute $\varepsilon''/\varepsilon' = \sigma/(\omega\varepsilon')$; classify medium
2. Find α, β (exact or approx from table)
3. $\eta_c = \eta_c e^{j\theta_\eta}$; $\tilde{\mathbf{H}} = \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{y}$
4. $\mathbf{H}(z, t) = \frac{ E_0 }{ \eta_c } e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \hat{y}$
5. $S_{av} = \frac{ E_0 ^2}{2 \eta_c } e^{-2\alpha z} \cos \theta_\eta \hat{z}$
6. $\delta_s = 1/\alpha$; $A[\text{dB}] = -8.69\alpha z$

η_c PHASE ANGLE QUICK-REF

Medium	θ_η	$\cos \theta_\eta$
Lossless	0	1
Low-loss diel.	≈ 0	≈ 1
Good cond.	$\approx \pi/4 = 45^\circ$	$\approx 1/\sqrt{2}$

Good cond.: $\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$; $|\eta_c| = R_s \sqrt{2}$.

LOSSY PARAMS

$\gamma = \alpha + j\beta$
α, β : exact from $\varepsilon''/\varepsilon'$
$\eta_c = \sqrt{\mu/\varepsilon_c} = \eta_c e^{j\theta_\eta}$
E, H phase diff = θ_η

GOOD COND.

$\delta_s = 1/\sqrt{\pi f \mu \sigma}$
$\tilde{J} = J_0 e^{-(1+j)z/\delta_s}$
$Z_s = R_s(1+j)$
$R_s = \sqrt{\pi f \mu/\sigma}$

AC RESISTANCE

$R_{ac} = R_s \ell/w$
$R'_{coax} = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$
$R_s \propto \sqrt{f}$
$P = \frac{1}{2} I ^2 R_s \ell/w$

READ LOSSLESS

$\eta = \eta_0 \sqrt{\mu_r/\varepsilon_r}$
$\tilde{\mathbf{H}} = (1/\eta) \hat{k} \times \tilde{\mathbf{E}}$
$S_{av} = E_0^2/(2\eta)$
$u_p = \omega/k = c/\sqrt{\mu_r \varepsilon_r}$

READ LOSSY

$\tilde{\mathbf{H}} = (E_0/\eta_c) e^{-\gamma z} \hat{y}$
$S_{av} \propto e^{-2\alpha z} \cos \theta_\eta$
$A = -8.69\alpha z$ (dB)
$\delta_s = 1/\alpha$